

White Paper 4 — Retrieving structures for the lead targets

Pipeline step: Retrieve structures for lead targets **Inputs:** lead list (STAT3, RORC, JAK3), UniProt accessions **Outputs:** `STAT3_6NJS.pdb`, `RORC_5APH.pdb`, `JAK3_5LWM.pdb`, `structure_inventory.csv`, `fig_lead_structures.png`

1. Purpose

Structure-based drug discovery needs an atomic-resolution model of the target's binding site. This step selects, for each lead, a high-quality experimental structure with a bound ligand — the ligand both marks the druggable pocket and provides a positive control for later docking.

2. Sources: UniProt and the PDB

Target identity was fixed through **UniProt**, the universal protein knowledgebase, which provides the canonical sequence and accession for each lead (STAT3 P40763, RORC P51449, JAK3 P52333) and the cross-references to structural entries [1]. Experimental structures were retrieved from the **Protein Data Bank (PDB)**, the global archive of macromolecular structures [2]. Searching the PDB by UniProt accession returns every deposited structure of the protein, which we then filtered by method, resolution, and ligand content.

3. Selection criteria

For each lead we preferred structures that were:

1. **X-ray, high resolution** (≤ 2.7 Å) — for reliable side-chain placement in the pocket;
2. **Ligand-bound** — a bound small molecule marks the orthosteric/functional site and anchors the docking box;
3. **Domain-appropriate** — covering the domain relevant to intervention (SH2/core for STAT3, the ligand-binding domain for RORyt, the kinase ATP site for JAK3).

Ligand identity was read directly from the structure's HETATM records and title rather than trusting a single metadata field, because ligand annotations in automated feeds are inconsistent.

4. Selected structures

Lead	PDB	Resolution	Domain	Reference ligand
STAT3	6NJS	2.70 Å	SH2 / core	KQV (degrader-class)
RORC / RORyt	5APH	1.54 Å	Ligand-binding domain (LBD)	VYI
JAK3	5LWM	1.55 Å	Kinase domain (ATP site)	79T

STAT3 has no crystal structure at the standard resolution cutoff under its primary accession; the search was broadened and 6NJS (the SH2/core in complex with a degrader-class compound) selected as the best available handle. RORyt and JAK3 both offered sub-1.6 Å ligand complexes — excellent starting points.

5. Downstream preparation

From each selected structure we extracted the reference ligand, computed the **pocket center** as the ligand centroid, and defined an initial docking box as the ligand bounding box padded by 16 Å. These pocket centers seed both the pocket-detection step (White Paper 5) and the docking box (White Paper 7). Where a full-length model is later needed for co-folding, **AlphaFold2** provides high-accuracy predicted structure as a complement to the experimental coordinates [3].

6. Deliverables

- `STAT3_6NJS.pdb`, `RORC_5APH.pdb`, `JAK3_5LWM.pdb` — the selected structures.
- `structure_inventory.csv` — all candidate structures with resolution, method, ligand, and domain.
- `fig_lead_structures.png` — Ca-ribbon overview of the three lead structures.

7. Caveats

- A single crystal form captures one conformational snapshot; pocket shape can differ in solution (addressed later by MD, White Paper 9).
- STAT3's best structure is a degrader complex, consistent with the difficulty of the classical SH2 pocket.

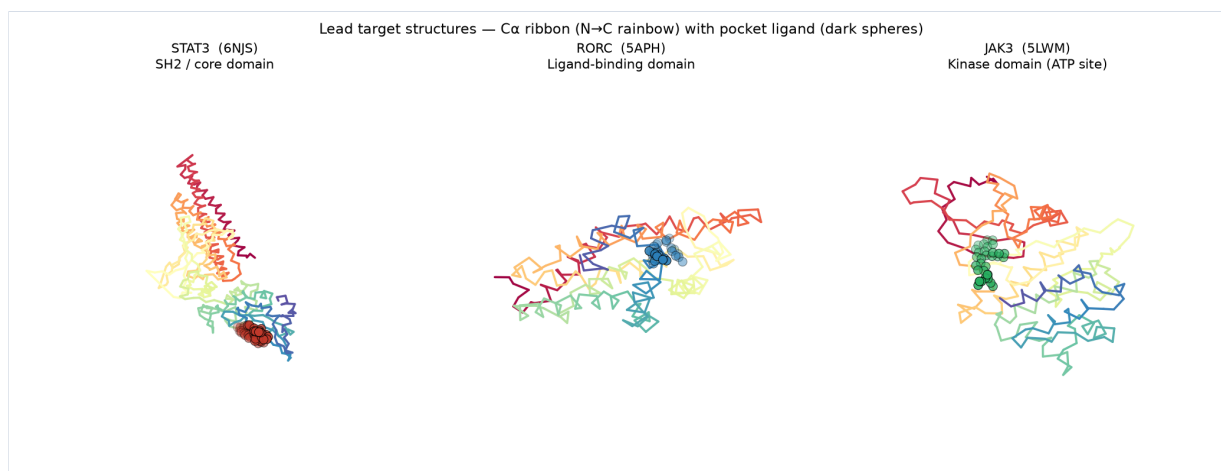


Figure: key output of this pipeline step.

References

1. The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*. 2022. doi:[10.1093/nar/gkac1052](https://doi.org/10.1093/nar/gkac1052)
2. Berman et al. The Protein Data Bank. *Nucleic Acids Research*. 2000. doi:[10.1093/nar/28.1.235](https://doi.org/10.1093/nar/28.1.235)
3. Jumper et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021. doi:[10.1038/s41586-021-03819-2](https://doi.org/10.1038/s41586-021-03819-2)